

Testing and Submission Checklist

Testing Approach

Testing is performed by running the application with representative road videos and visually checking the output window and saved output video. Each test verifies that the expected stage of the pipeline completes without error and produces a meaningful result.

Test Cases

Test ID	Input / Test Case	Expected Output	Observed Output
TC-01	Valid data/input/test_video.mp4 with clear lane markings	Video opens, lane lines are overlaid, and output MP4 is saved	Lane lines are displayed and data/output/output_video.mp4 is created.
TC-02	Frame containing visible left and right lane boundaries	Lanes are identified separately with different colours	Left lane is blue and right lane is red when markings are clear.
TC-03	Video with small frame-to-frame changes	Lines should not flicker severely	Averaging and temporal smoothing make lines more stable.
TC-04	Press Q or Esc during display	Application exits safely and output written so far is saved	Window closes and video resources are released safely.
TC-05	Missing data/input/test_video.mp4	Clear error message instead of an application crash	Program reports the missing input-video path.
TC-06	Faded or partially hidden lane markings	Detection may be weaker, but program continues running	Fewer or unstable lines may appear; application remains operational.

Manual Verification Steps

1. Confirm that data/input/test_video.mp4 exists.
2. Install dependencies with `pip install -r requirements.txt`.
3. Run `python src/main.py` from the project root.
4. Verify that the output window opens and FPS text is visible.
5. Verify that the detected left and right lanes appear in blue and red.
6. Open data/output/output_video.mp4 in a media player after processing.
7. Confirm that the saved video plays and contains lane overlays.

GitHub Submission Checklist

Files to Upload

- [] `src/main.py`
- [] `src/utils.py`
- [] `requirements.txt`
- [] `README.md`
- [] `statement.md`
- [] `docs/REPORT_CONTENT.md`
- [] `docs/DIAGRAMS.md`
- [] `docs/TESTING_AND_SUBMISSION.md`
- [] Screenshots in `docs/screenshots/` (recommended)
- [] A small sample input video only if allowed by the course or repository-size limit

Before Pushing

- [] Confirm the folder structure matches the README.
- [] Confirm `requirements.txt` contains `opencv-python` and `numpy`.
- [] Run the project once using the command in the README.
- [] Confirm `data/output/output_video.mp4` is generated and playable.
- [] Add screenshots to the README placeholders.
- [] Do not upload unnecessary cache folders such as `__pycache__`.
- [] Add a `.gitignore` file if working with Git, especially for generated videos and cache files.

Suggested Git Commands

```
git init
git add src requirements.txt README.md statement.md docs
git commit -m "Add real-time lane detection project"
git branch -M main
git remote add origin <your-github-repository-url>
git push -u origin main
```